

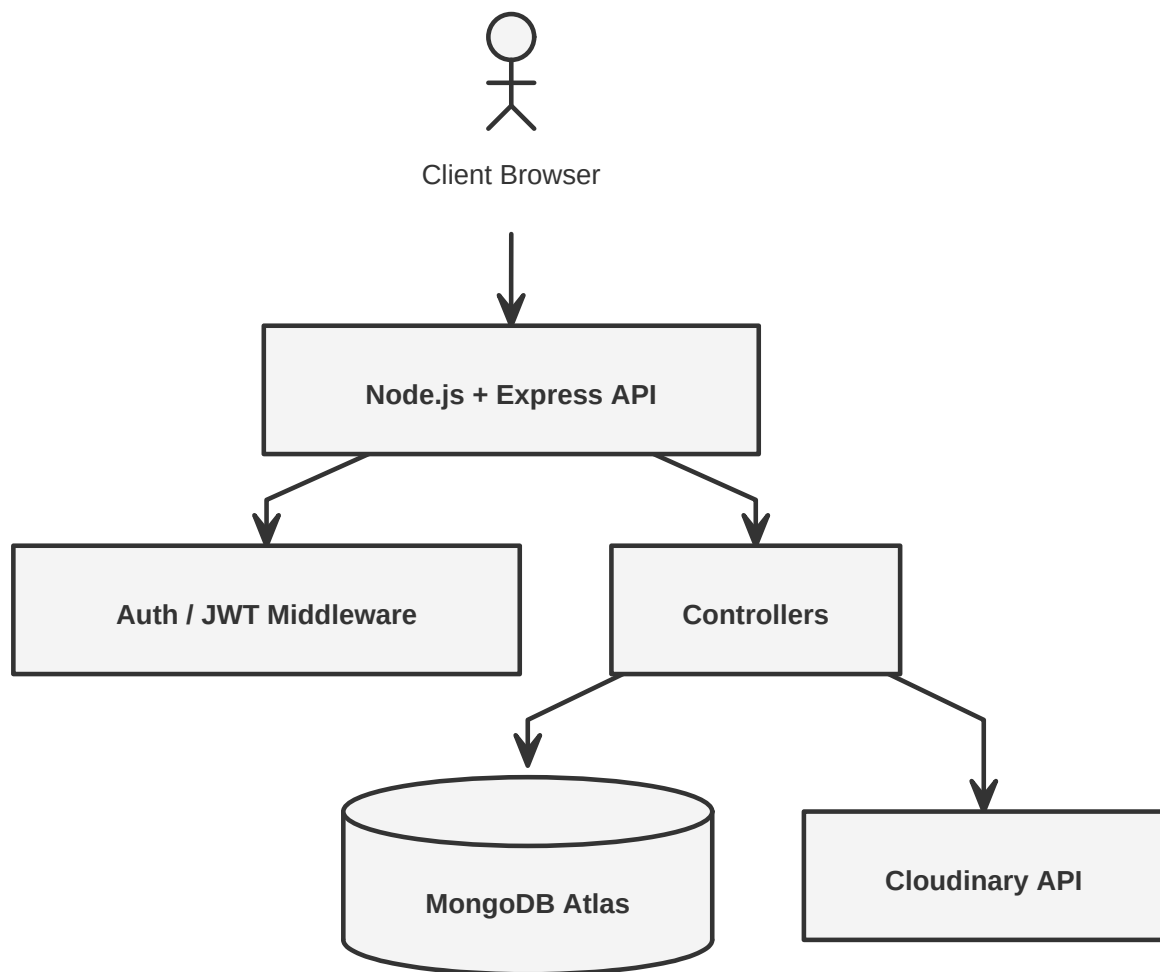
Solution Architecture

Date: 11 June 2026

Project Name: E-commerce Application

Architecture Overview

Architecture Diagram



The system follows a conceptual decoupling of client-server application flows using MongoDB, Express, React, and Node.js. It adheres strictly to the classic Model-View-Controller (MVC) design pattern mapped across the codebase.

1. Frontend (View)

- **Technology:** React.js
- **Description:** A responsive Single Page Application (SPA) providing an interactive user interface.
- **Key Responsibilities:** State management (handling cart items, user session), routing, displaying dynamic components (product lists, reviews, checkout steps).

2. Backend (Controller)

- **Technology:** Node.js & Express.js
- **Description:** A RESTful API that handles business logic and acts as the bridge between the frontend and database.
- **Key Responsibilities:** User authentication (Bcrypt, JWT), routing API requests, payment gateway integration (Payment Gateway), processing orders, and handling Cloudinary uploads for images.

3. Database (Model)

- **Technology:** MongoDB (NoSQL) with Mongoose ODM
- **Description:** A document-oriented database storing JSON-like documents.
- **Key Entities:**
 - **User:** Credentials, role, address.
 - **Product:** Details, pricing, stock, reviews.
 - **Order:** Shipping, payment details, order items.
 - **Category:** Product categorizations.
 - **Cart:** Temporary/persistent cart state.

4. Third-Party Integrations

- **Payment:** Payment API (Planned).
- **Storage:** Cloudinary API for image hosting and manipulation.